So what does this have to do with deep reinforcement learning? Well, Q-learning, as you've learned, gives us the expected (average, time-discounted) state-action values. As you might imagine, this can lead to the same limitations we've been discussing in the case of drugs, with multimodal distributions. Learning a full probability distribution of state-action values would give us a lot more power than just learning the expected value, as in ordinary Q-learning. With the full distribution, we could see if there is multimodality in the state-action values and how much variance there is in the distribution. Figure 7.4 models the action-value distributions for three different actions, and you can see that some actions have more variance than others. With this additional information, we can employ risk-sensitive policies—policies that aim not merely to maximize expected rewards but also to control the amount of risk we take in doing so.

Figure 7.4. Top: An ordinary Q function takes a state-action pair and computes the associated Q value. Middle: A distributional Q function takes a state-action pair and computes a probability distribution over all possible Q values. Probabilities are bounded in the interval [0,1], so it returns a vector with all elements in [0,1] and their sum is 1. Bottom: An example Q value distribution produced by the distributional Q function for three different actions for some state. Action A is likely to lead to an average reward of –5, whereas action B is likely to lead to an average reward of +4.